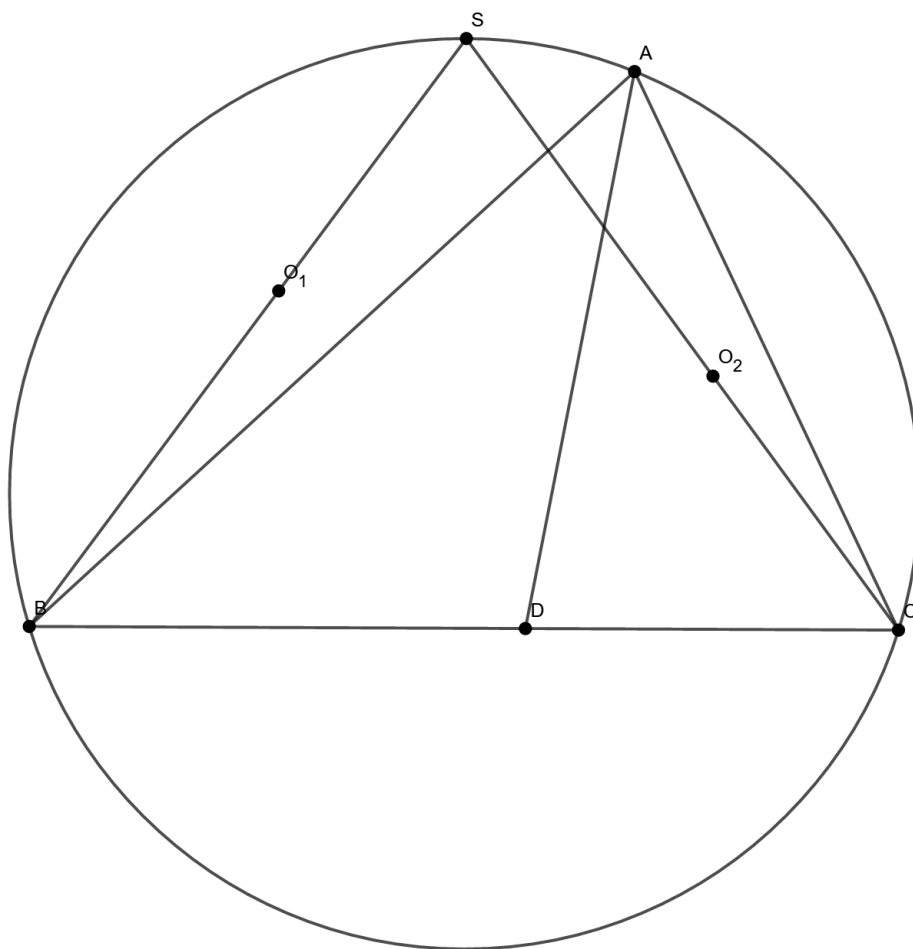


4.3.22



$\angle SBC = 90^\circ - \frac{\angle BAC}{2} = 90^\circ - \angle BAD = \angle O_1BD$, so B, O_1, S are collinear. Similarly, C, O_2, S are collinear.